

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – MODEL COMPARISON PRESENTATION (BERT VS GPT VS T5 VS LLAMA)

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO4 – Precision (BL3, BL4,BL5)	Assessment:	Assessment Tool 2 - Model Comparison Presentation (BERT vs GPT vs T5 vs LLaMA)
Weightage:	10% of CIA	Total Marks:	2 * 50= 100
CO4 Statement:	<i>Design and implement multimodal Generative AI applications integrating text, image, and speech models for engineering applications.(BL3, BL4,BL5)</i>		
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Section / Batch:		Date:	

Instructions to Students

- **Answer any 2 questions. Each question carries 50 marks. Total: 100 marks.**
- **Compare all four models: BERT, GPT, T5, and LLaMA.**
- Explain the **architecture and working principle** of each model.
- Compare their **pre-training objectives and capabilities**.
- Map each model to appropriate **real-world applications**.
- Identify the **strengths and limitations** of each model.
- Support the presentation using **credible research papers, official documentation, benchmarks, or other reliable sources**.
- Include at least one **comparison table or diagram**.
- Use relevant **real-world case studies or application examples**.
- Conclude by identifying the **most suitable model for the selected application** and justify the decision.
- Include proper **references/citations** in the presentation.

Code of Conduct

I, S. Muskan(192472182), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **S. Muskan**

SECTION A : MODEL COMPARISON PRESENTATION (BERT VS GPT VS T5 VS LLAMA)

ARCHITECTURE-BASED COMPARISON

2. Compare encoder-only, decoder-only, and encoder-decoder architectures using BERT, GPT, and T5 as representative models. Explain where LLaMA fits and justify your classification.

36. A college wants to build a RAG-based document question-answering system using academic regulations and course documents. Compare BERT, GPT, T5, and LLaMA and recommend a suitable model.

Q2. Architecture-Based Comparison of BERT, GPT, T5 and LLaMA

Introduction

BERT, GPT, T5 and LLaMA are Transformer-based language models, but they use different architectures and training objectives.

- **BERT** → **Encoder-only**
- **GPT** → **Decoder-only**
- **T5** → **Encoder-decoder**
- **LLaMA** → **Decoder-only**

The architecture determines how each model processes information and which NLP tasks it is best suited for.

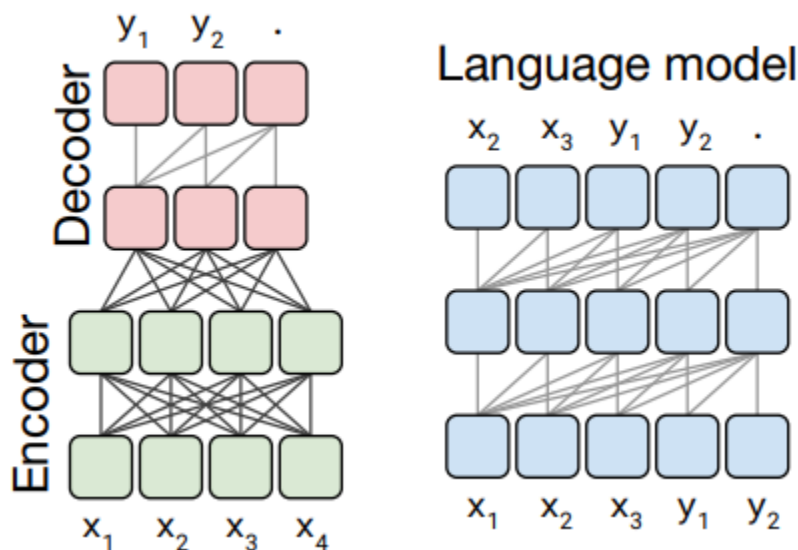


Diagram showing Encoder-only, Decoder-only and Encoder-decoder architectures

BERT – Encoder-Only Architecture

Architecture:

BERT uses only the **Transformer encoder**. It uses bidirectional self-attention, allowing each token to use information from both left and right context.

Working:

Input text → Tokenization → Encoder layers → Contextual representation → Task output.

Pre-training objective:

BERT was originally trained using **Masked Language Modeling (MLM)** and **Next Sentence Prediction (NSP)** [1].

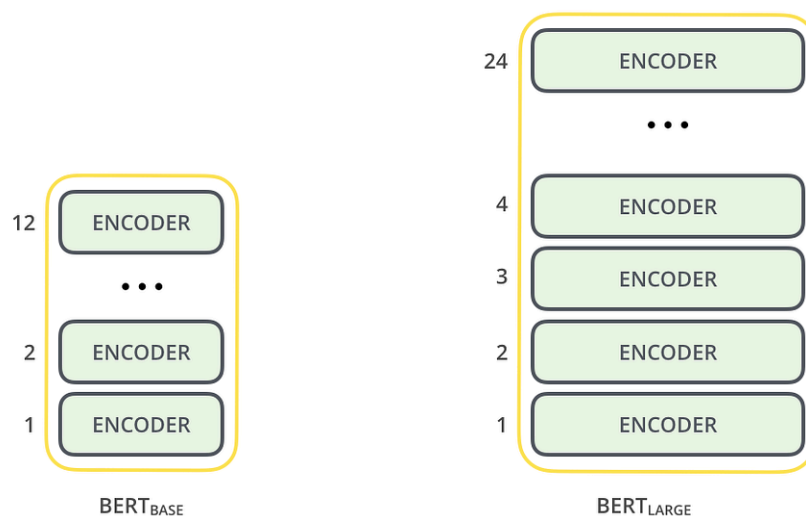
Capabilities:

- Text classification
- Sentiment analysis
- Named Entity Recognition
- Semantic similarity
- Extractive Question Answering

Strength: Excellent contextual language understanding.

Limitation: Not designed for open-ended text generation.

Applications: Search, document classification, sentiment analysis and information extraction.



BERT encoder-only architecture

GPT – Decoder-Only Architecture**Architecture:**

GPT uses a **Transformer decoder-only architecture** with causal self-attention.

Working:

Input prompt → Decoder layers → Predict next token → Generate next token → Repeat.

Pre-training objective:

GPT uses **Causal Language Modeling**, predicting the next token from previous tokens [2].

Capabilities:

- Text generation
- Question answering
- Summarization
- Chatbots
- Code generation
- Few-shot learning

Strength: Strong and flexible text generation.

Limitation: Can produce incorrect or unsupported information and large models can require significant computation.

Applications: Chatbots, content generation, coding assistants and AI assistants.

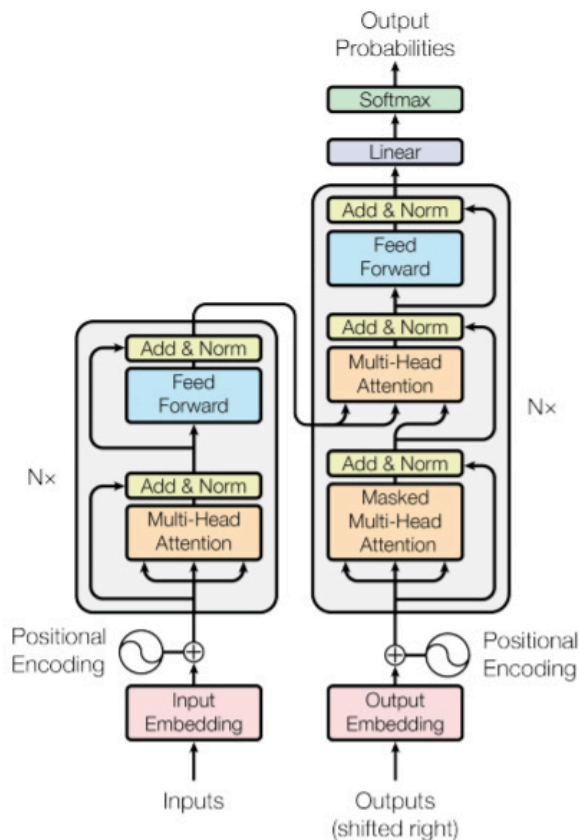


Figure 1: The Transformer - model architecture.

GPT decoder-only architecture with causal attention

T5 – Encoder-Decoder Architecture

Architecture:

T5 contains both a **Transformer encoder and decoder**.

Working:

Input text → Encoder → Context representation → Decoder → Output text.

Pre-training objective:

T5 uses a **text-denoising/span-corruption objective** and treats NLP tasks as **text-to-text problems** [3].

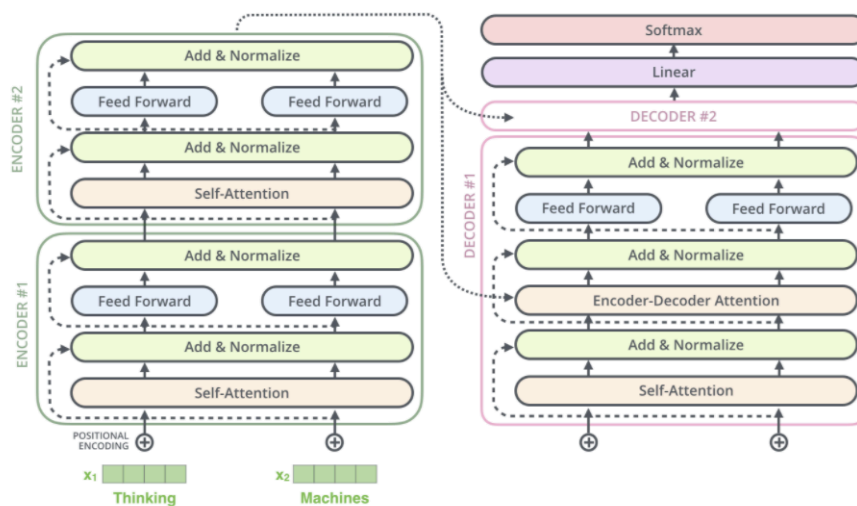
Capabilities:

- Translation
- Summarization
- Question answering
- Classification
- Text transformation

Strength: One unified architecture can handle many NLP tasks as text-to-text problems.

Limitation: Larger models require considerable computational resources.

Applications: Translation systems, summarization and question-answering systems.



T5 encoder-decoder architecture

LLaMA – Decoder-Only Architecture

Architecture:

LLaMA uses a **decoder-only Transformer architecture**, placing it in the same broad architectural category as GPT [4].

Working:

Input prompt → Decoder layers → Causal self-attention → Next-token prediction → Generated response.

Pre-training objective:

LLaMA uses **autoregressive/causal language modeling**, predicting the next token [4].

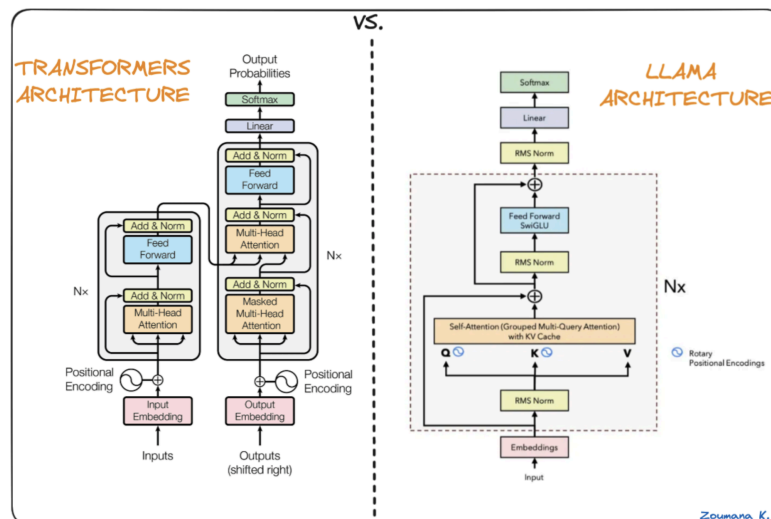
Capabilities:

- Text generation
- Question answering
- Dialogue
- Code generation
- Instruction-based applications
- RAG-based generation

Strength: Strong generative capability with opportunities for customization and controlled deployment.

Limitation: Larger models require significant memory and computation, and generated responses may contain errors.

Applications: AI assistants, chatbots, RAG systems and customized language applications.



LLaMA decoder-only architecture

Architecture Comparison

Feature	BERT	GPT	T5	LLaMA
Architecture	Encoder-only	Decoder-only	Encoder-decoder	Decoder-only
Main focus	Understanding	Generation	Text-to-text	Generation
Attention	Bidirectional	Causal	Encoder + decoder	Causal
Pre-training	MLM + NSP*	Causal LM	Span corruption	Causal LM
Text generation	Limited	Excellent	Excellent	Excellent
Classification	Excellent	Good	Good	Good
Summarization	Limited	Excellent	Excellent	Excellent
Translation	Not primary	Possible	Excellent	Possible
Chatbot	Not primary	Excellent	Good	Excellent
Code generation	Not primary	Strong	Possible	Strong

*NSP refers to BERT's original pre-training formulation.

Input and Output Processing

BERT

Input → Encoder → Representation → Classification/Extraction

GPT

Prompt → Decoder → Next-token prediction → Generated text

T5

Input text → Encoder → Decoder → Output text

LLaMA

Prompt → Decoder → Next-token prediction → Generated response

[IMAGE PLACEHOLDER 6: Four-model input/output processing flow]

Pre-Training Objective Comparison

Model	Objective	Main Effect
BERT	Masked Language Modeling	Learns bidirectional representations
GPT	Causal Language Modeling	Learns sequential text generation
T5	Span corruption / denoising	Learns text-to-text transformation
LLaMA	Causal Language Modeling	Learns autoregressive generation

The training objective strongly influences the model's capabilities. BERT's objective emphasizes understanding, while GPT and LLaMA's causal objective naturally supports generation. T5's encoder-decoder objective supports transforming one text sequence into another.

Real-World Application Mapping

Application	Most Suitable Model	Reason
Sentiment analysis	BERT	Strong contextual classification
Named Entity Recognition	BERT	Strong language understanding
Text generation	GPT / LLaMA	Decoder-based generation
Summarization	T5 / GPT / LLaMA	Strong text generation/transformation
Translation	T5	Encoder-decoder sequence transformation
Chatbot	GPT / LLaMA	Natural conversational generation
Extractive QA	BERT	Strong text understanding
Generative QA	GPT / T5 / LLaMA	Can generate complete answers
Code generation	GPT / LLaMA	Strong autoregressive generation

Conclusion

BERT, GPT, T5 and LLaMA demonstrate three major Transformer architectures.

- **BERT** is best suited for understanding and extracting information.
- **GPT** is designed for flexible autoregressive generation.
- **T5** is suitable for text-to-text tasks such as translation and summarization.
- **LLaMA** is a decoder-only generative model suitable for assistants, chatbots and RAG applications.

Therefore, the architecture should be selected according to the application's requirements rather than assuming that one model is best for every task.

References

[Q2-1] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. Proceedings of NAACL-HLT 2019.

[Q2-2] Brown, T. B., et al. (2020). *Language Models are Few-Shot Learners*. Advances in Neural Information Processing Systems.

[Q2-3] Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., & Liu, P. J. (2020). *Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer*. Journal of Machine Learning Research, 21(140), 1–67.

[Q2-4] Touvron, H., et al. (2023). *LLaMA: Open and Efficient Foundation Language Models*. arXiv:2302.13971.

[Q2-5] Touvron, H., et al. (2023). *Llama 2: Open Foundation and Fine-Tuned Chat Models*. arXiv:2307.09288.

[Q2-6] Vaswani, A., et al. (2017). *Attention Is All You Need*. Advances in Neural Information Processing Systems, 30.

Q36. RAG-Based College Document Question-Answering System

Case Study

A college wants to build a question-answering system using:

- Academic regulations
- Course documents
- Examination rules
- Attendance policies
- Student handbooks
- Syllabus documents

Students could ask questions such as:

"What is the minimum attendance required?"

Instead of manually searching documents, the college can use **Retrieval-Augmented Generation (RAG)**.

RAG System

RAG combines document retrieval with a language model.

Basic workflow

College Documents

↓

Document Processing & Chunking

↓

Embeddings

↓

Vector Database

↓

User Question

↓

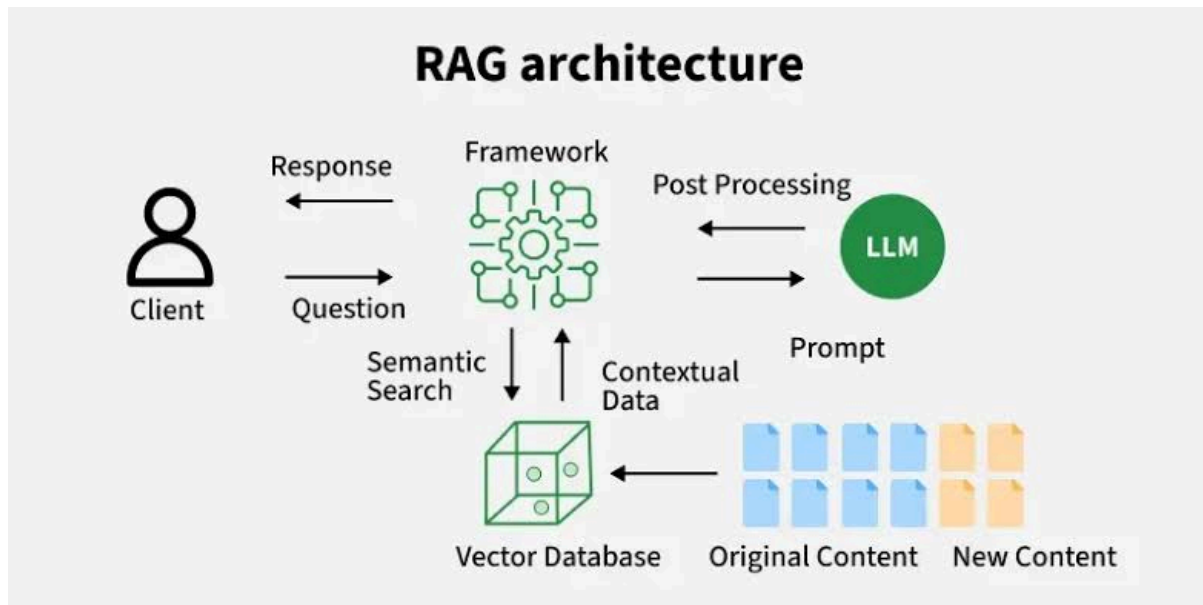
Retrieve Relevant Documents

↓

Language Model

↓

Grounded Answer



Complete RAG architecture for the college system

Why RAG is Required

A language model's pretrained knowledge may not contain the college's latest regulations.

For example, if the college document states:

"Students must maintain 80% attendance."

A RAG system retrieves this information before generating the answer.

Therefore, the system can provide answers based on the **college's actual documents** rather than relying only on the model's general knowledge.

Role of BERT in the RAG System

BERT can be used mainly for the **understanding/retrieval side**.

Possible roles

- Question understanding
- Semantic representation
- Document classification
- Relevance matching
- Extractive question answering
- Reranking

Limitation

BERT is not designed for open-ended conversational answer generation.

Best role: Retrieval/understanding component.

Role of GPT in the RAG System

GPT can be used as the **answer-generation component**.

Workflow

Student Question → Retrieval → Relevant Context → GPT → Answer

GPT can:

- Generate natural-language answers
- Summarize retrieved regulations
- Answer student questions
- Support conversational interaction

Best role: RAG answer generator.

Role of T5 in the RAG System

T5 can use the retrieved information as input and generate an answer.

Workflow

Question + Retrieved Context → T5 → Answer

It is particularly suitable for:

- Question answering
- Summarization
- Text transformation

Best role: Text-to-text answer generation.

Role of LLaMA in the RAG System

LLaMA can act as the **primary answer-generation model**.

Workflow

Student Question → Retrieval → Relevant Academic Context → LLaMA → Answer

It is suitable for:

- Conversational responses
- Question answering
- Summarization
- Customized assistants
- RAG applications

Best role: Customizable RAG answer generator.

Four-Model RAG Comparison

Model	RAG Role	Strength	Limitation
BERT	Retrieval/understanding	Strong contextual representations	Not designed for generation

GPT	Answer generation	Strong natural-language generation	Deployment/cost depends on model/service
T5	Answer generation	Strong text-to-text processing	Larger models can require significant resources
LLaMA	Answer generation	Generation + customization	Requires suitable compute for deployment

Example College RAG Interaction

Student Question

"What percentage attendance is required?"

Retrieved Context

"Students must maintain a minimum attendance of 80%."

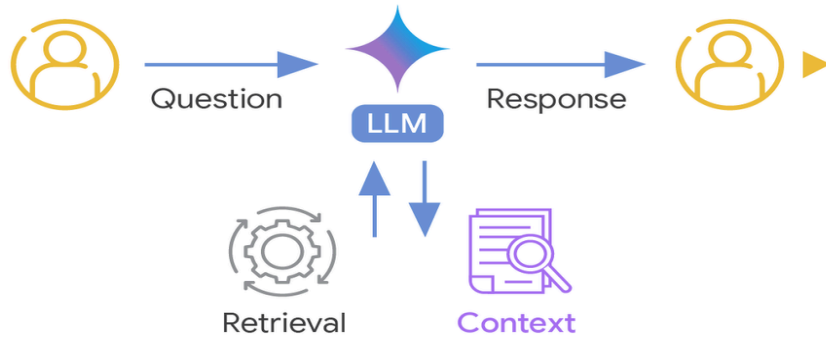
Generated Answer

"According to the academic regulation, students must maintain a minimum attendance of 80%."

Source

Academic Regulations – Attendance Section

This approach makes the answer **grounded in the college document**.



Example showing Question → Retrieved Context → Generated Answer

Other Real-World Questions

The system could answer questions such as:

Examination

"What happens if I miss the end-semester examination?"

Course

"How many credits are required to complete the program?"

Attendance

"What is the minimum attendance requirement?"

Regulations

"What are the eligibility requirements for the examination?"

The RAG system retrieves the appropriate document section before generating the answer.

Strengths and Limitations for the College Application

Model	Strengths	Limitations
BERT	Excellent understanding and retrieval support	Weak for conversational generation
GPT	Strong generation and conversation	May require external service and can hallucinate
T5	Strong QA, summarization and text transformation	Computational requirements for larger models
LLaMA	Strong generation, customization and deployment control	Requires suitable compute and grounding

Recommended RAG Architecture

For the college system, the recommended architecture is:

College Documents



Chunking



Embedding Model



Vector Database



Retriever



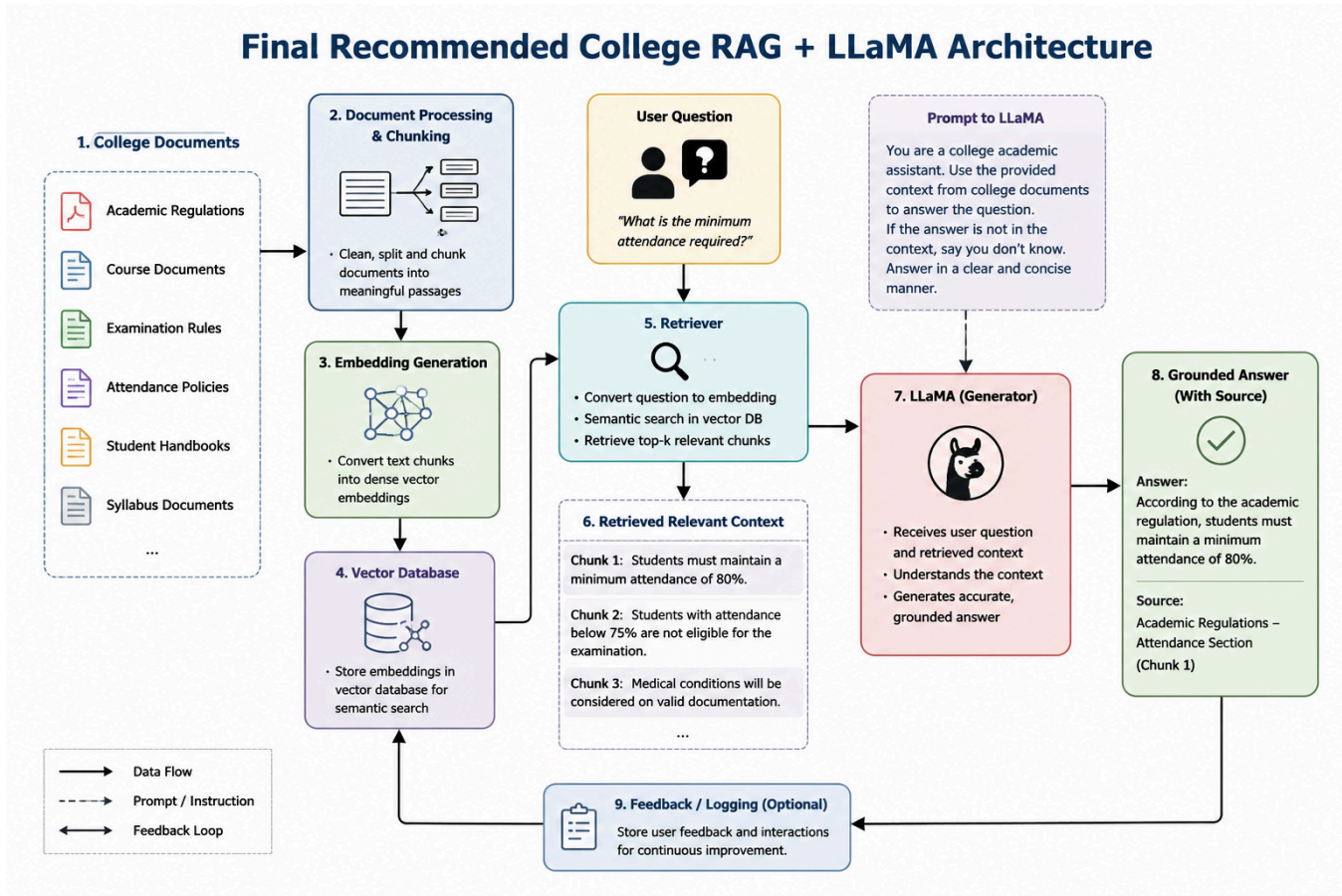
Relevant Context



LLaMA



Grounded Answer



Final recommended College RAG + LLaMA architecture

Why LLaMA is Recommended

LLaMA is selected as the primary generation model because:

- 1. It uses a decoder-only architecture suitable for generation.
- 2. It can produce conversational answers.
- 3. It can be integrated with retrieved academic documents.
- 4. It offers opportunities for customization.
- 5. Suitable model sizes can be selected according to deployment resources.
- 6. It can support a college-specific AI assistant.

However, **GPT is also a strong alternative** when the college prefers a managed model/service and prioritizes ease of deployment.

Final Comparison

Requirement	BERT	GPT	T5	LLaMA
Understand query	Excellent	Excellent	Excellent	Excellent
Retrieval support	Excellent	Good	Good	Good
Generate answers	Limited	Excellent	Excellent	Excellent
Summarization	Good	Excellent	Excellent	Excellent
Conversational QA	Limited	Excellent	Good	Excellent
Customization	Strong	Depends on service	Strong	Strong
RAG generation	Limited	Excellent	Excellent	Excellent

Recommended role	Retrieval	Generator	Generator	Primary Generator
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Conclusion

A RAG-based college document QA system can combine document retrieval with a language model to provide answers grounded in academic regulations and course documents.

BERT is most useful for **understanding and retrieval-related tasks**, while GPT, T5 and LLaMA can generate answers.

For this application, **LLaMA is recommended as the primary generation model** because it combines strong generative capability with customization and deployment flexibility.

The final system is:

College Documents → Retrieval → Relevant Context → LLaMA → Grounded Answer

This approach provides a practical AI assistant while reducing dependence on the model's pretrained knowledge.

Final Conclusion

The comparison shows that **architecture and training objective determine the strengths of each model**.

BERT is most appropriate when deep language understanding is required.

GPT is highly suitable for general-purpose text generation and conversational applications.

T5 is particularly suitable for text-to-text tasks such as translation, summarization and question answering.

LLaMA is suitable for generative applications where customization and deployment flexibility are important.

For the selected Q36 case study, the recommended solution is:

RAG + LLaMA

with a retrieval system providing relevant college documents as context.

The final system should also provide **document/source references** with answers so students can verify the information.

References

- [1] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. NAACL-HLT.
- [2] Brown, T. B., et al. (2020). *Language Models are Few-Shot Learners*. Advances in Neural Information Processing Systems.
- [3] Raffel, C., et al. (2020). *Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer*. Journal of Machine Learning Research, 21(140), 1–67.
- [4] Touvron, H., et al. (2023). *LLaMA: Open and Efficient Foundation Language Models*. arXiv:2302.13971.
- [5] Touvron, H., et al. (2023). *Llama 2: Open Foundation and Fine-Tuned Chat Models*. arXiv:2307.09288.
- [6] Lewis, P., et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. Advances in Neural Information Processing System.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Architectural Comparison Accuracy	30%	Accurate, in-depth comparison of architectures and use-cases	Accurate comparison with minor gaps	Superficial comparison of models	Inaccurate or confused comparison
Application Mapping	25%	Clearly maps each model to appropriate real-world applications	Reasonable mapping to applications	Weak mapping to applications	No meaningful mapping presented
Research Depth	20%	Well-researched with credible sources/benchmarks cited	Adequately researched	Limited research depth	Unresearched / opinion-based content
Delivery	15%	Confident, engaging, well-paced delivery	Clear delivery with minor pacing issues	Hesitant delivery	Unclear or unprepared delivery
Slide Design & References	10%	Well-designed slides with proper references	Adequate slides, minor reference gaps	Basic slides, poor referencing	No slides or references provided

Marks Evaluation:

Question No.	Architectural Comparison Accuracy (30%)	Application Mapping (25%)	Research Depth (20%)	Delivery (15%)	Slide Design & References (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						